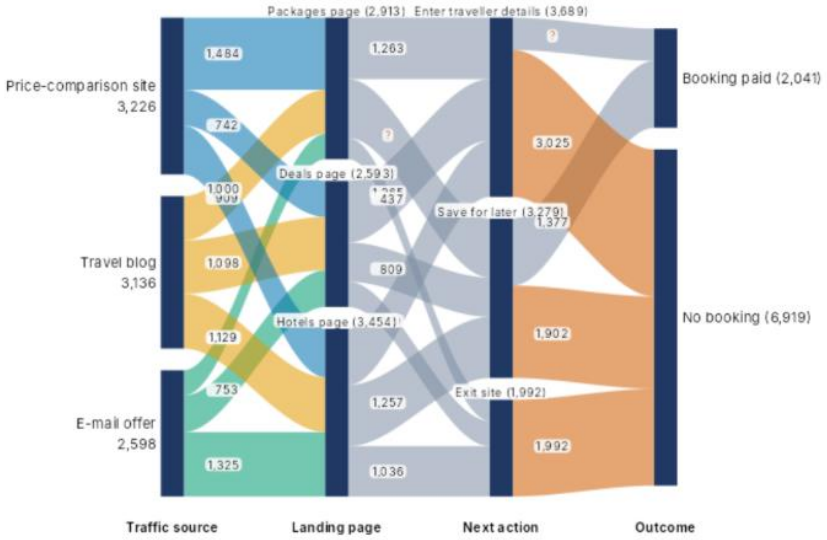
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot D

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Scenario: The analytics team of a travel-booking website tracked 8,960 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Packages page to Save for later; Hotels page to Enter traveller details; Enter traveller details to Booking paid. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Booking paid as a percentage of all sessions; (ii) the largest single flow into "No booking" or "Exit site"; (iii) the percentage of "Deals page" sessions that go on to "Enter traveller details". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 400 sessions. Compute the height in cm of each landing-page node (Packages page, Deals page, Hotels page). [2]

(e) Write R code: a links data frame for the first four flows leaving "E-mail offer" and "Price-comparison site" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

Aim

To analyze website visitor flow using a Sankey diagram, calculate hidden flows and conversion rates, and understand how data visualization helps identify user behavior.

(a) Find the Three Hidden Flows Using Conservation of Flow

The conservation of flow principle states that the total flow entering a node must equal the total flow leaving it.

1. Packages page → Save for later

Total sessions on Packages page = 2,913

Known flow to Enter traveller details = 1,263

Hidden flow = 2,913 − 1,263 = **1,650 sessions**

2. Hotels page → Enter traveller details

Total sessions entering Enter traveller details = 3,689

Known incoming flows:

- Packages page = 1,263
- Save for later = 1,377

Hidden flow = 3,689 − 1,263 − 1,377 = **1,049 sessions**

3. Enter traveller details → Booking paid

Total sessions entering Enter traveller details = 3,689

Sessions going to No booking = 3,025

Hidden flow = 3,689 − 3,025 = **664 sessions**

Therefore, the calculated hidden flows are 1,650, 1,049, and 664 sessions, respectively.

(b) Calculations

(i) Overall conversion rate

Total sessions = 8,960

Booking paid = 2,041

Formula:

Conversion rate = (Booking paid / Total sessions) × 100

= (2,041 / 8,960) × 100

= **22.78%**

(ii) Largest single flow into No booking or Exit site

- Flow into No booking = 3,025 sessions
- Flow into Exit site = 1,992 sessions

Therefore, the largest single flow is **3,025 sessions into No booking**.

(iii) Percentage of Deals page sessions going to Enter traveller details

Deals page sessions = 2,593

Flow to Enter traveller details = 437

Percentage = (437 / 2,593) × 100

= **16.85%**

(c) Four Design Decisions for the Final Sankey Diagram

1. **Node order:** Arrange traffic source, landing page, next action, and outcome from left to right. This makes the website journey easy to follow.
2. **Colour rule:** Use different colours for traffic sources and consistent colours for outcomes. This helps distinguish the flows clearly.
3. **Labels:** Display node names and session counts. This helps users understand the data without confusion.
4. **Highlight important flows:** Highlight paid bookings and the largest flow into No booking. This helps identify successful conversions and areas where users leave without booking.

(d) Calculate the Height of Each Landing-Page Node

Given: 1 cm of node height represents 400 sessions.

Formula:

Node height = Number of sessions / 400

1. Packages page

Height = 2,913 / 400 = **7.28 cm**

2. Deals page

Height = 2,593 / 400 = **6.48 cm**

3. Hotels page

Height = 3,454 / 400 = **8.64 cm**

(e) R Code to Create a Sankey Diagram

```
install.packages("networkD3")

library(networkD3)


# Create nodes

nodes <- data.frame(

 name = c(

  "E-mail offer",

  "Price-comparison site",

  "Packages page",
```

```
   "Deals page",
   "Hotels page"
  )
)

# Create the first four flows
links <- data.frame(
  source = c(0, 0, 1, 1),
  target = c(2, 3, 2, 3),
  value = c(753, 1325, 1484, 742)
)

# Draw the Sankey diagram
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 25
)
```

The nodes data frame contains the names of the categories, while the links data frame stores the source, target, and flow values. The sankeyNetwork() function creates the visualization.

(f) Two Situations Where a Sankey Diagram Can Be Misleading or Unreadable

1. Too many nodes and crossing flows: The diagram becomes difficult to read when it contains many categories and overlapping links. A bar chart is a suitable alternative.

2. Very small flows: Small flows may be difficult to identify when compared with large flows. A data table can be used to display exact values clearly.

Result

The website session flow was analyzed using a Sankey diagram. The overall booking conversion rate was calculated as 22.78%, and the largest flow into No booking was identified as 3,025 sessions. The landing-page node heights were calculated, and suitable design decisions and alternative charts were identified.